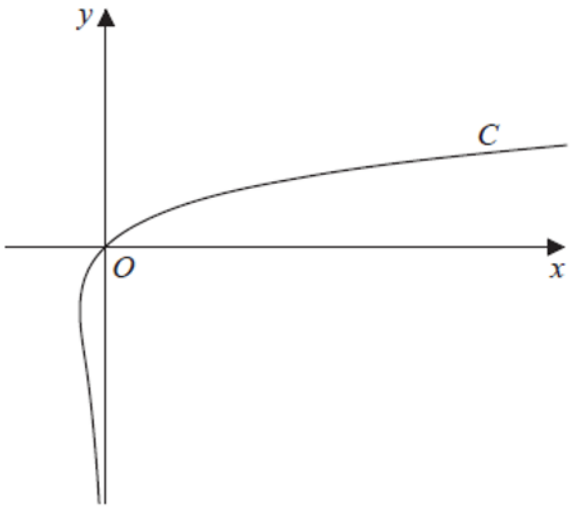
Questions

Q1.



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Figure 3

Figure 3 shows a sketch of the curve C with equation

$$x = ye^{2y} \qquad y \in \mathbb{R}$$

(a) Show that

$$\frac{dy}{dx} = \frac{y}{x(1+2y)}$$

(4)

Given that the straight line with equation $x = k$, where k is a constant, cuts C at exactly two points,

(b) find the range of possible values for k.

(3)

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(Total for question = 7 marks)

(Q10 WMA13/01, Jan 2022)

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Q2.

Find, using calculus, the x coordinate of the stationary point on the curve with equation

$$y = (2x + 5)e^{3x}$$

(Total for question = 4 marks)

(Q01 WMA13/01, Jan 2022)

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Q3.

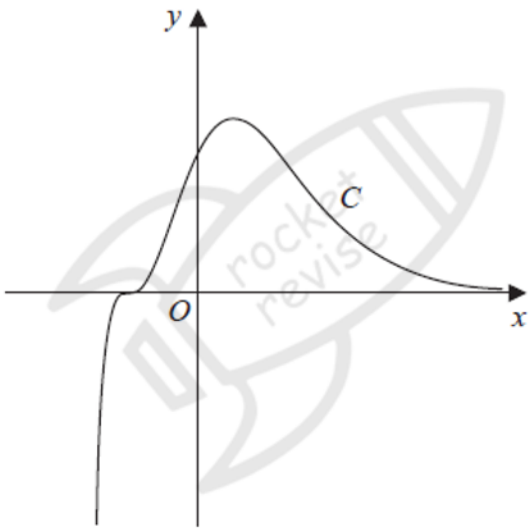


Figure 3

Figure 3 shows a sketch of the curve C with equation $y = f(x)$, where

$$f(x) = (2x + 1)^3e^{-4x}$$

(a) Show that

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$$f'(x) = A(2x + 1)^2 (1 - 4x) e^{-4x}$$

where A is a constant to be found.

(4)

(b) Hence find the exact coordinates of the two stationary points on C.

(3)

The function g is defined by

$$g(x) = 8f(x - 2)$$

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(c) Find the coordinates of the maximum stationary point on the curve with equation $y = g(x)$.

(2)

(Total for question = 9 marks)

(Q08 WMA13/01, June 2023)

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Q4.

$$y = \log_{10}(2x + 1)$$

(a) Express x in terms of y.

(2)

(b) Hence, giving your answer in terms of x, find $\frac{dy}{dx}$

(3)

(Total for question = 5 marks)

(Q04 WMA13/01, Oct 2022)

Q5.

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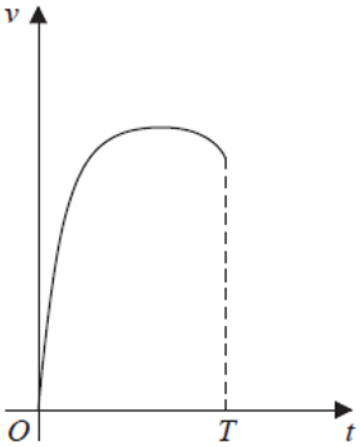


Figure 4

Figure 4 is a graph showing the velocity of a sprinter during a 100 m race.

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The sprinter's velocity during the race, v m s^{-1}, is modelled by the equation

$$v = 12 - e^{t-10} - 12e^{-0.75t} \qquad t \geq 0$$

where t seconds is the time after the sprinter begins to run.

According to the model,

(a) find, using calculus, the sprinter's maximum velocity during the race.

(5)

Given that the sprinter runs 100 m in T seconds, such that

$$\int_0^T v \, dt = 100$$

(b) show that T is a solution of the equation

$$T = \frac{1}{12}(116 - 16e^{-0.75T} + e^{T-10} - e^{-10})$$

(4)

The iteration formula

$$T_{n+1} = \frac{1}{12}(116 - 16e^{-0.75T_n} + e^{T_n-10} - e^{-10})$$

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is used to find an approximate value for T

Using this iteration formula with $T_1 = 10$

(c) find, to 4 decimal places,

(i) the value of T_2

(ii) the time taken by the sprinter to run the race, according to the model.

(3)

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(Total for question = 12 marks)

(Q08 WMA13/01, June 2022)

Q6.

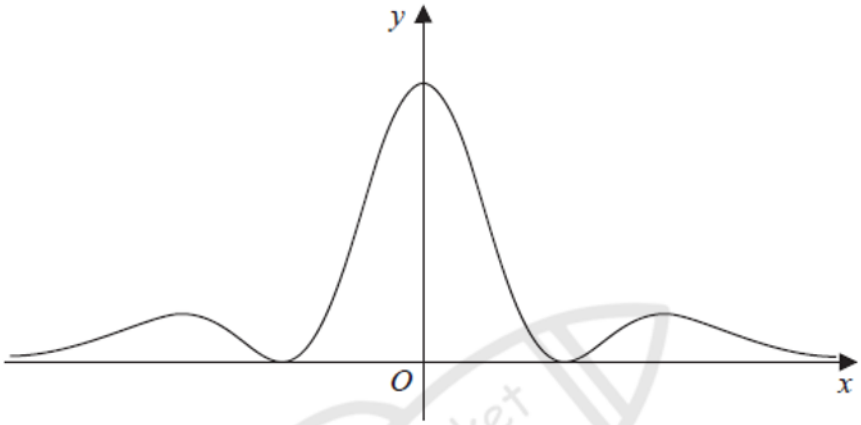


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \mathrm{e}^{-x^2}(2x^2 - 3)^2$$

(a) Find the range of f

(2)

(b) Show that

$$\mathrm{f}'(x) = 2x(2x^2 - 3)\mathrm{e}^{-x^2}(A - Bx^2)$$

where A and B are constants to be found.

(4)

Given that the line $y = k$, where k is a constant, $k > 0$, intersects the curve at exactly two distinct points,

(c) find the exact range of values of k

(4)

(Total for question = 10 marks)

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(Q07 WMA13/01, Oct 2023)

Q7.

The curve C has equation

$$y = \frac{\ln(x^2 + k)}{x^2 + k} \qquad x \in \mathbb{R}$$

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where k is a positive constant.

(a) Show that

$$\frac{dy}{dx} = \frac{Ax(B - \ln(x^2 + k))}{(x^2 + k)^2}$$

where A and B are constants to be found.

(3)

Given that C has exactly three turning points,

(b) find the x coordinate of each of these points. Give your answer in terms of k where appropriate.

(3)

(c) find the upper limit to the value for k.

(1)

(Total for question = 7 marks)

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(Q05 WMA13/01, Oct 2023)

Q8.

A population of a rare species of toad is being studied.

The number of toads, N, in the population, t years after the start of the study, is modelled by the

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equation

$$N = \frac{900e^{0.12t}}{2e^{0.12t} + 1} \qquad t \geqslant 0, t \in \mathbb{R}$$

According to this model,

(a) calculate the number of toads in the population at the start of the study,

(1)

(b) find the value of t when there are 420 toads in the population, giving your answer to 2 decimal places.

(4)

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(c) Explain why, according to this model, the number of toads in the population can never reach 500

(1)

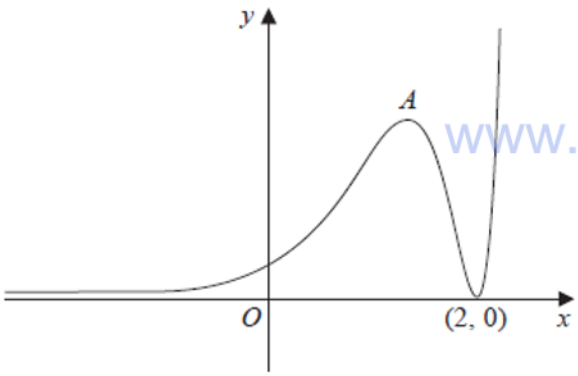
(Total for question = 6 marks)

(Q01 WMA13/01, Jan 2020)

Q9.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.



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Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = f(x)$ where

$$f(x) = (x - 2)^2 e^{3x} \qquad x \in \mathbb{R}$$

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The curve has a maximum turning point at A and a minimum turning point at (2, 0)

(a) Use calculus to find the exact coordinates of A.

(5)

Given that the equation f(x) = k , where k is a constant, has **at least** two distinct roots,

(b) state the range of possible values for k.

(2)

(Total for question = 7 marks)

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(**Q03 WMA13/01, Oct 2022**)

Q10.

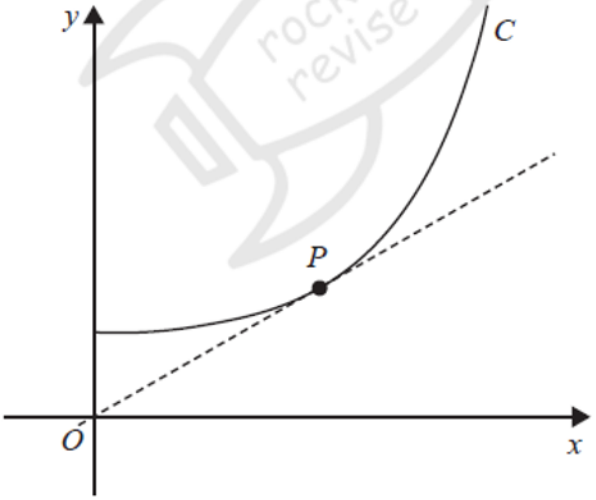


Figure 3

Figure 3 shows a sketch of part of the curve C with equation

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$$y = \sqrt{3 + 4e^{x^2}} \qquad x \geqslant 0$$

(a) Find $\frac{dy}{dx}$, giving your answer in simplest form.

(2)

The point P with x coordinate α lies on C.

Given that the tangent to C at P passes through the origin, as shown in Figure 3,

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(b) show that $x = \alpha$ is a solution of the equation

$$4x^2e^{x^2} - 4e^{x^2} - 3 = 0$$

(3)

(c) Hence show that α lies between 1 and 2

(2)

(d) Show that the equation in part (b) can be written in the form

$$x = \frac{1}{2}\sqrt{4 + 3e^{-x^2}}$$

(1)

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The iteration formula

$$x_{n+1} = \frac{1}{2}\sqrt{4 + 3e^{-x_n^2}}$$

with $x_1 = 1$ is used to find an approximation for α.

(e) Use the iteration formula to find, to 4 decimal places, the value of

(i) x^3

(ii) α

(3)

(Total for question = 11 marks)

(Q09 WMA13/01, Jan 2023)

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Q11.

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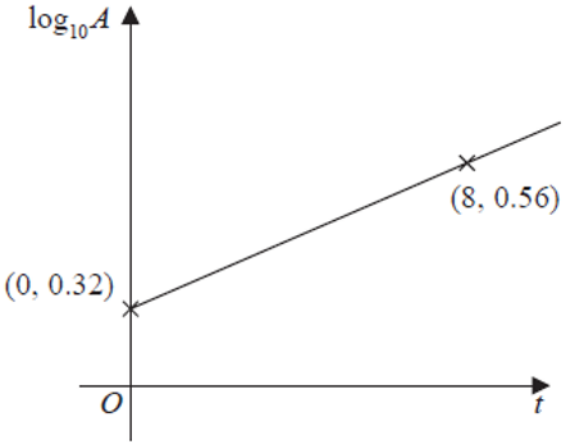


Figure 1

The growth of duckweed on a pond is being studied.

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The surface area of the pond covered by duckweed, $A\text{m}^2$, at a time t days after the start of the study is modelled by the equation

$$A = pq^{t} \qquad \text{where } p \text{ and } q \text{ are positive constants}$$

Figure 1 shows the linear relationship between $\log_{10} A$ and t.

The points (0, 0.32) and (8, 0.56) lie on the line as shown.

(a) Find, to 3 decimal places, the value of p and the value of q.

(4)

Using the model with the values of p and q found in part (a),

(b) find the rate of increase of the surface area of the pond covered by duckweed, in m^2 / day, exactly 6 days after the start
of the study.
Give your answer to 2 decimal places.

(3)

(Total for question = 7 marks)

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(Q05 WMA13/01, June 2021)

Q12.

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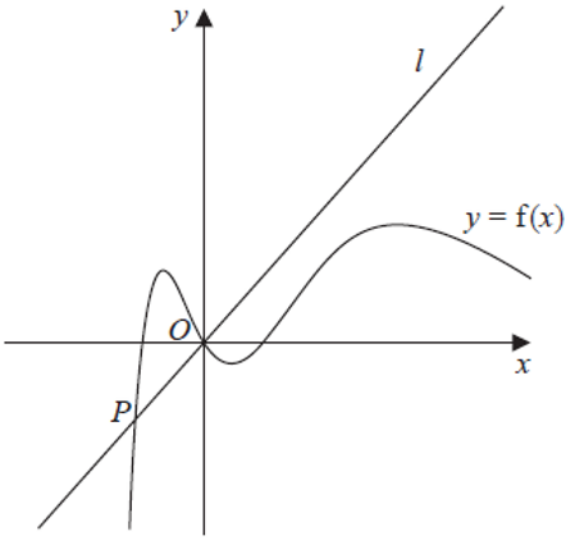


Figure 3

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Figure 3 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = x(x^2 - 4)e^{-\frac{1}{2}x}$$

(a) Find $f'(x)$.

(2)

The line l is the normal to the curve at O and meets the curve again at the point P.

The point P lies in the 3rd quadrant, as shown in Figure 3.

(b) Show that the x coordinate of P is a solution of the equation

$$x = -\frac{1}{2}\sqrt{16 + e^{\frac{1}{2}x}}$$

(4)

(c) Using the iterative formula

$$x_{n+1} = -\frac{1}{2}\sqrt{16 + e^{\frac{1}{2}x_n}} \quad \text{with } x_1 = -2$$

find, to 4 decimal places,

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(i) the value of x_2

(ii) the x coordinate of P.

(3)

(Total for question = 9 marks)

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